One Plot, Two Reporting Strategies

AP Statistics · Topic 2.4 (CED 5.1) · B: 2027-02-05 · E: 2027-03-31 · TEACHER KEY

CED TETHER

- **Skill 2.B** — Construct numerical or graphical representations of distributions.
- **Skill 2.A** — Describe data presented numerically or graphically.
- **EU UNC-1** — Graphical representations and statistics allow us to identify and represent key features of data.
- **LO UNC-1.S** — Represent bivariate quantitative data using scatterplots. [Skill 2.B]
- **EK UNC-1.S.1** — A bivariate quantitative data set consists of observations of two different quantitative variables made on individuals in a sample or population.
- **EK UNC-1.S.2** — A scatterplot shows two numeric values for each observation, one corresponding to the value on the x-axis and one corresponding to the value on the y-axis.
- **EK UNC-1.S.3** — An explanatory variable is a variable whose values are used to explain or predict corresponding values for the response variable.
- **EU DAT-1** — Regression models may allow us to predict responses to changes in an explanatory variable.
- **LO DAT-1.A** — Describe the characteristics of a scatter plot. [Skill 2.A]
- **EK DAT-1.A.1** — A description of a scatter plot includes form, direction, strength, and unusual features.
- **EK DAT-1.A.2** — The direction of the association shown in a scatterplot, if any, can be described as positive or negative.
- **EK DAT-1.A.3** — A positive association means that as values of one variable increase, the values of the other variable tend to increase. A negative association means that as values of one variable increase, values of the other variable tend to decrease.
- **EK DAT-1.A.4** — The form of the association shown in a scatterplot, if any, can be described as linear or non-linear to varying degrees.
- **EK DAT-1.A.5** — The strength of the association is how closely the individual points follow a specific pattern, e.g., linear, and can be shown in a scatterplot. Strength can be described as strong, moderate, or weak.
- **EK DAT-1.A.6** — Unusual features of a scatter plot include clusters of points or points with relatively large discrepancies between the value of the response variable and a predicted value for the response variable.

Essential question: How should several scatterplot features be weighed when descriptions conflict?

DOK-3 skill: 4.B · **FRQ pattern:** `linear-label-vs-curved-pattern-description-and-consequence`

DOK rationale: Part (c) requires coordinating direction, changing rate, strength, and an unusual observation to adjudicate two reporting strategies and trace a practical consequence.

Do Now → video follow-along → finish and turn in **DOK:** 1 → 3 **Minutes:** 45

Teacher does

- Project the first take; require one cited feature.
- Start the 2.4 follow-along at minute 5.
- At minute 33, require students to reconcile the rate changes and 50-minute value.

Students do

- Commit to a reporting strategy, then complete the follow-along.
- Finish (a)–(c), revise the first take, and turn in.

Questions to ask

- What do the early and late changes imply about form?
- How unusual is the 50-minute value locally?
- Which strategy preserves both facts?

Adult role
Circulate; withhold confirmation until students cite form and the unusual point.

[WARN] WATCH FOR

- Equating positive direction with linear form.
- Letting one point determine the entire pattern.
- Omitting context or strength.

SCORING PART (c) — E / P / I

Expected elements

- **selects-strategy** (required) — Selects Ravi and gives a complete contextual description
- **uses-changing-rate** (required) — Uses 170 mL versus 15 mL to justify nonlinear form
- **uses-unusual-point** (required) — Uses the 122.5 mL local discrepancy while preserving direction and strength
- **traces-consequence** (required) — Explains how a straight-line account could distort a later-growth decision

E	Selects Ravi; coordinates direction, form, strength, both changes, the unusual point, and a concrete consequence.
P	Makes the warranted choice but incompletely links one feature or consequence.
I	Equates positive with linear, lets one point erase the pattern, or gives an unsupported choice.

Common mistakes

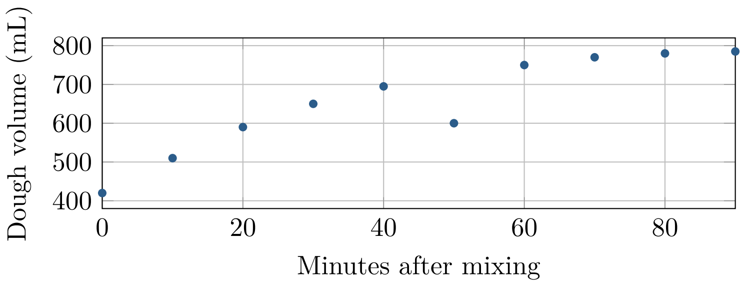
- Calling every positive association linear
- Letting one point determine the entire form
- Omitting context, strength, or the practical consequence

[STAR] TODAY'S PROBLEM — annotated

Suppose a test kitchen measures dough volume for ten batches at assigned times after mixing. The scatterplot shows the paired measurements.

Jules: Use a straight-line summary because volume generally increases with time. Treat the 50-minute batch as ordinary variation.

Ravi: Decide the form from how the rate changes, and report the 50-minute batch separately if it departs from nearby values.



FIRST TAKE (5 min)

Which reporting strategy seems more defensible, Jules's or Ravi's? Cite one graph feature.
Key: Not graded; students revise after coordinating all features.

Rules callout "BUILD A COMPLETE SCATTERPLOT DESCRIPTION (keep on the page)" is printed on the student sheet.

DOK 1 **(a)** Identify the explanatory and response variables, assign them to axes, and interpret $(30, 650)$.

Key: Time after mixing is the explanatory variable on the x-axis; dough volume is the response on the y-axis. $(30, 650)$ is a batch measured at 650 mL after 30 minutes.

DOK 2 **(b)** Find the volume changes from 0 to 20 and 70 to 90 minutes. Compare the 50-minute volume with the midpoint of the 40- and 60-minute volumes.

Key: The changes are $590 - 420 = 170$ mL and $785 - 770 = 15$ mL. The local midpoint is $(695 + 750)/2 = 722.5$ mL, so 600 mL is 122.5 mL below it.

DOK 3 **(c)** Choose a reporting strategy. Use direction, form, strength, both changes, and the 50-minute comparison to write a complete description. Explain how the rejected strategy could distort a later-growth decision.

Key: Ravi's strategy is warranted. The association is fairly strong, positive, and nonlinear: most points rise but level off, with a 170-mL early increase versus only 15 mL late. The 50-minute batch is unusually low, 122.5 mL below its local benchmark, and should be reported without erasing the nine-point curve. Jules's straight-line account could overstate later growth and treat the low batch as typical.

EXIT

One thing the video changed about my first take: ______________________________